Discussion Point:

How often do you think that open licence software is being used? Should it be used more often?

TASK 10.01

Carry out a search to investigate some of the software available under an open licence.

KEY TERMS

Open source software: software provided with the source code

Free software: software provided with the source code when the user is free to use it as they wish

Shareware: software provided free of charge for a limited period but no source code provided

Freeware: software provided free of charge with no time limit for its use but no source code provided

10.06 Artificial intelligence (AI)

Artificial intelligence (AI) depends on and draws from many other disciplines including: philosophy, psychology, neuroscience, mathematics, linguistics and control engineering. The only definitions of AI that are acceptable are at the same time so generalised that they are not very practical. The following is a typical example:

AI concerns the use of a computer or computer-controlled device to perform tasks normally associated with intelligent behaviour by humans.

We will consider five aspects of intelligent human behaviour and discuss some applications of AI that mimic this human behaviour.

Problem solving

One example is the development of a system that can play chess. This can be considered as displaying artificial intelligence but this is only demonstrated because the rules of chess are limited. A computer with sufficient storage capacity and processing power can investigate so many options for a possible sequence of moves that a human cannot compete.

A second example is the traditional form of expert system that, for example, has been developed to aid medical diagnosis. This is supplied with data and rules from living medical experts. The expert system contains more knowledge than is possible for an individual doctor to have. However, if the expert system is given a new situation that is not covered by the data and rules it has been given, it cannot attempt a new or creative approach – unlike a human.

Linguistics

Voice recognition and voice synthesis techniques are already developed and in use. One example is if you phone a help line where you might be answered by a computer. Provided that you answer questions clearly the computer might be able to identify your needs and pass you on to an appropriate human who can help. However, this is a long way away from the computer itself creating new questions based on your answers and providing the help you need.

Perception

Traditionally robots have been used in manufacturing processes. Here the robot is programmed to perform repetitive tasks. The action of the robot each time is triggered by some mechanism. However, if anything unexpected happens the robot continues to operate as normal, regardless of any damage being caused.

There is now much research focussed on the development of autonomous robots. These have to be fitted with sensors to enable the robot to take appropriate action depending on the information received from the sensors. This is an example of perception in AI.

A development of this concept is the driverless car. There are several examples available or in development but so far they have only been able to perform limited tasks. An example is the capability for a car to park itself in a vacant parking space.

Reasoning

There are examples of the application of AI where a program has been able to draw inferences (reach conclusions based on evidence) which is a requirement for reasoning. The best examples concern the proving of mathematical theorems. Attempts have also been made to develop techniques that can verify that software that has been created does indeed correctly and fully match the documented specification.

Learning

This is currently a very active area for the application of AI techniques. Machine learning is said to take place if a system that has a task to perform is seen to improve its performance as it gains experience. The AI system has access to 'experience' in the form of a massive set of data. By the use of appropriate statistical algorithms the system learns from this data.

One example is when the actions of users visiting websites to buy products are stored. The AI system then attempts to identify appropriate products to be advertised when a user returns to the website. If sales progressively increase there is evidence that learning is taking place.

Another example is the program that investigates incoming emails and makes decisions as to whether these can be classified as spam and therefore should be refused entry to the user inbox.

The impact of AI

The use of the Internet dominates the lives of a large proportion of the world's population. Global organisations that provide the systems underpinning this user activity are collecting and storing massive amounts of data concerning how the Internet is being used. If this data is only being used to enable the organisation to increase its profits, this could be seen as normal business practice. However, if the data is not being securely stored it could get into the wrong hands and be used for criminal or subversive activity.

There are different concerns with respect to the introduction of autonomous mechanical products such as robots, robotic devices and driverless vehicles into our daily lives. There are arguments that technological developments lead to employment of more people to manufacture, service and install the new products. There is a further argument that more technology leads to less manual labour and therefore to increased leisure time. One counter argument is that more technology leads to fewer jobs because machines are doing the work. Another is that such developments simply make the rich richer and the poor poorer.

Some people are excited by the introduction of driverless vehicles, but other people believe that the potential for accidents will be increased and that there are not enough measures to prevent accidents.

Robots can be used in environments that would be dangerous for humans to enter. Giving the robot the capability to act autonomously would make it more useful in such environments.

The environmental impact of robot manufacture and disposal is probably the most significant issue. Robots are manufactured and require materials for their construction. There is only a limited supply of the raw materials needed. Also, all mechanical and electronic devices eventually end up on the scrap heap contributing to the already serious problem of waste products harming the environment and creatures living in this environment.

The use of improved expert systems to aid practising doctors and nurses is clearly a benefit. However, if these systems came to replace doctors and nurses the social consequences are difficult to predict.

Discussion Point:

Have you seen any recent information about new developments in AI?

Reflection Point:

Is there an organisation in your country for professional computer scientists? If so, does it encourage young people to join?

Summary

- There are different definitions of ethics.
- Professional organisations provide rules of conduct that include guidance on ethical behaviour.
- There is a history of software disasters that might have been prevented if good software engineering practice had been employed.
- Copyright is formal recognition of ownership.
- The use of software is controlled by a license
- Only open source software is provided with the source code which allows freedom of usage
- Artificial Intelligence is currently mainly focused on the development of autonomous mechanical products and machine learning based on access to massive data sets.

Exam-style Questions

1 a Complete the following sentences:

i As a computer professional your primary concern when faced with an issue should be .....

ii If an issue arises you should exercise your ..... and possibly seek .....

iii You have a responsibility to act in accordance with the welfare of ....

iv You are expected to act in the interests of your ..... and of your .....

v You should not accept ..... for which you lack ..... [8]

b Explain **two** reasons why documentation is mentioned so often in the ACM/IEEE code of conduct. [4]

2 **a** Copyright is an important consideration when something is created.

i State what copyright primarily defines. [1]

ii When copyright is registered, some data will be recorded. Identify **two** examples of the type of data that would be recorded. [2]

iii Copyright legislation defines two conditions that will apply to the copyrighted work. Identify **one** of these. [1]

iv When copyright has been established there are options for how usage will be controlled. Give **two** alternatives for the instructions that could be included in the copyright statement for the created item. [2]

b When software is obtained there will be an associated licence defining how it can be used.

i For commercial software, describe **two** different ways in which the licence might be applied and explain the benefits to the customer of one of these. [4]

ii Define the difference between freeware and shareware. [2]

3 Identify **two** applications of artificial intelligence. For each one identify an aspect of human intelligence that the application mimics. Either explain how the application will be a benefit or explain why there would be concern about its use. [8]

4 Bobby is a senior programmer at a software house which produces intruder detection software. He also runs his own software company which develops and sells various computer applications.

The following table shows seven activities which Bobby carries out.

Put a tick (✓) in the appropriate column to identify if the activity is ethical or unethical.

Activity	Ethical	Unethical
Gives away passwords used in the intruder detection software		
Uses source code developed at the software house for the software he develops for his own company		
Insists that staff work to deadlines		
Turns down training opportunities offered by his employer		
Writes and sells software that reads confidential data from client computers		
Fakes test results of safety-critical software		
Has the software applications developed overseas for sale in his own country		

[7]

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5 A team of software engineers is developing a new e-commerce program for a client.

State **three** of the principles of the ACM/IEEE Software Engineering Code of Ethics. Illustrate each one, with an example, describing how it will influence their working practices. [6]

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